Sources of lifetime inequality revisited

Carlos Lizama

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Conference Opportunities, Mobility and Well-being

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- Initial conditions are somewhat determined by parental background.
- Important for insurance purposes.
- Many ways initial conditions matter. Focus on one particular mechanism: human capital accumulation.
- How much initial conditions matter? Which conditions have the greatest quantitative impact?
- How much does parental background/wealth influence in determining the initial conditions and lifetime welfare?

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- Intergenerational transmission of wealth can alleviate the borrowing constraints for wealthier households.
- There is also well documented intergenerational transmission of abilities/ human capital, which also matters for determining lifetime welfare.

This paper

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- Quantitatively assess the importance of initial conditions: human capital, ability and initial wealth.
- Assess the importance of parental background and transfers, and their implication for intergenerational mobility.

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- Initial wealth inequality affects lifetime welfare and income. Accounting for intergenerational correlation of abilities seems less important.
- This results is in stark contrast with other studies. Taking into account intergenerational wealth transfers and costly human capital is key.

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- **Human capital and borrowing constraints:** Lochner, Monge-Maranjo (2011), Hai and Heckman (2017)

Outline

The model

Empirical Analysis

Calibration

Results and Discussion

Conclusions

The model

Demographics, preferences and heterogeneity

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- Heterogeneity: Agents are heterogeneous in three aspects: (i) innate ability θ, (ii) financial wealth a and human capital h. Ability is fixed throughout lifetime, whereas agents can accumulate financial wealth and human capital.

Life cycle

- Education stage

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- Retirement
  - No labor earnings. They live off a pension benefit from the government and their wealth. They do not leave bequests.

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- Human capital production function $h_{t+1} = \exp(z_{t+1})(h_t(1-\delta) + \theta(s_t h_t)^{\alpha})$

The government and borrowing constraints

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- Two different borrowing restrictions: no borrowing and ad-hoc credit constraints.

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- Parents transfer wealth to their kids and they can make the same decisions about schooling.
- Allow for intergenerational correlation of ability and initial human capital. At this point, they are uncorrelated and initial ability and human capital are drawn from distribution $g(h, \theta)$

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- Retire at age 65.

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- They have a kid at age 27.
- At age 50, kid turns 23 and parent decides how much to transfers.
- Retire at age 65.
- Die at 80.

Bellman equations

The Bellman equation for a worker who decides whether to go to college is

$$V(a, h; \theta) = \max\{V^c(a, h; \theta); V^{nc}(a, h; \theta)\}$$

The Bellman equation for a worker hired at wage w

$$\begin{aligned} V_j(a_j, h_j; \theta) &= \max_{c_j, s_j} u(c_j) + \beta \mathbb{E}_z[V_{j+1}(a_{j+1}, h_{j+1}; \theta)] \\ \text{s.t.} \qquad c_j + a_{j+1} &= a_j + \lambda(ra_j + (1 - \tau_{ss})wh_j(1 - s_j))^{1-\tau} \\ h_{j+1} &= \exp(z_{t+1})((1 - \delta)h_j + \theta(s_j h_j)^{\alpha}) \end{aligned}$$

A college student pays at the beginning of life P_c to go to college. This is paid with their initial endowment given by parents.

Bellman equations (2)

The Bellman equation of a parent who decides how much wealth to transfer to his offspring is

$$V(a_j, h_j; \theta) = \max_{c_j, s_j, a_j^*} u(c) + \beta \mathbb{E}[V(a_{j+1}, h_{j+1}; \theta)] + \gamma \mathbb{E}\left(V^o(a_j^*, h_o; \theta_o)\right)$$

$$\text{s.t.} \qquad c_j + a_{j+1} = a_j + \lambda(ra_j + (1-\tau_{ss})wh_j(1-s_j))^{1-\tau} - a_j^*$$

$$h_{j+1} = \exp(z_{t+1})((1-\delta)h_j + \theta(s_j h_j)^\alpha)$$

The Bellman equation for a retiree is

$$V(a_j) = \max_{c_j} + \beta V(a_{j+1})$$

$$\text{s.t.} \qquad c_j + a_{j+1} = p + (1+r)a_j$$

Equilibrium

The equilibrium in this economy is a sequence of policy functions $\{c_j, s_j\}_{j=1}^{J}$, value functions $\{V_j\}_{j=1}^{J}$ and measures over states $\{g(a, h; \theta)\}_{j=1}^{J}$ such that, given prices r, w_l, w_h, the government policy and the initial distribution $g_0(a, h; \theta)$:

- Policy functions and Value functions solve all Value Functions.
- Measures $\{g(a, h; \theta)\}_{j=1}^{J}$ are consistent with policy functions and shocks.

Empirical Analysis

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- Data sources: PSID, NLSY79, NLSY97.
- Income profiles and inequality measure in the life cycle for a cohort.
- Income profiles by education level.
- Intergenerational persistence of education.
- Education and wealth.
- Education and parental transfers.

Data

- Panel Study of Income Dynamics
  - First interviews in 1968. It samples more than 18,000 individuals from 5,000 families in the US. Representative of the US. Annual waves until 1997 and biannual since then.
  - Detailed data for household head and wife, some details about kids.
  - Once kids leave home and form their own household unit, they are kept in the survey as a separate household unit.
  - I use these data to compute statistics throughout lifetime.
- National Longitud Survey of Youth
  - Two cohorts, 1979 and 1997 (NLSY79, NLSY97). Approximately 12.000 and 9.000 individuals surveys in their first rounds, respectively.
  - Surveys youth between 14 and 22 years old in the first round. It follows those individuals since then.
  - I use these data to compute income profiles and compute intergenerational transfers from parents.

Income profiles and inequality measures over the life cycle

- Focus on male, household heads from PSID.
- Each age is a bin of 5 years with center in age.
- I compute stat_j as a weighted average of stat, weighted by the number of observations in time t and cohort h.

Income profile over the life cycle

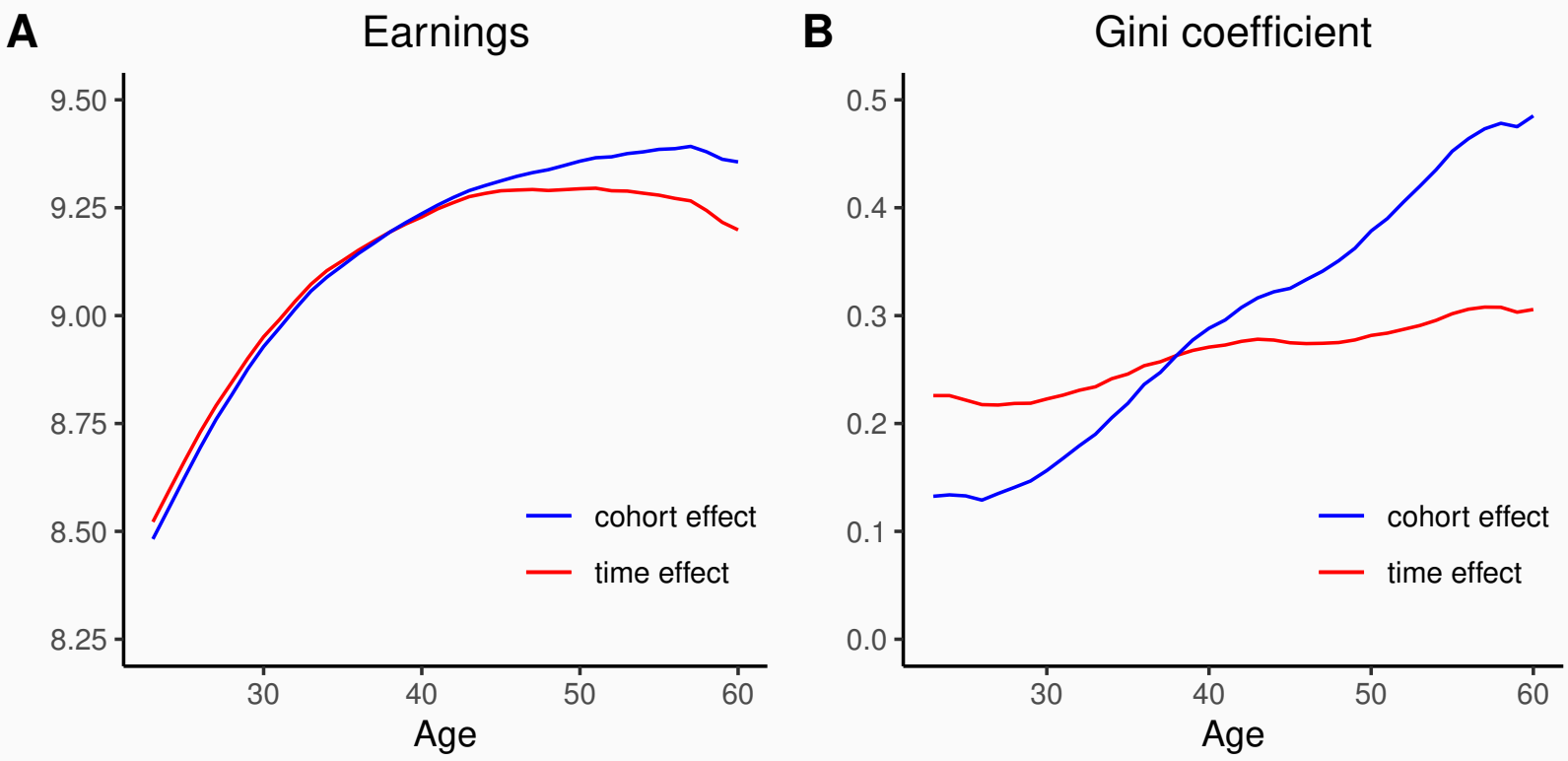

Figure 1: mean earnings and Gini by age

Income profile by education level: PSID

- Analyze two separate sub samples. Non-college and college educated people.

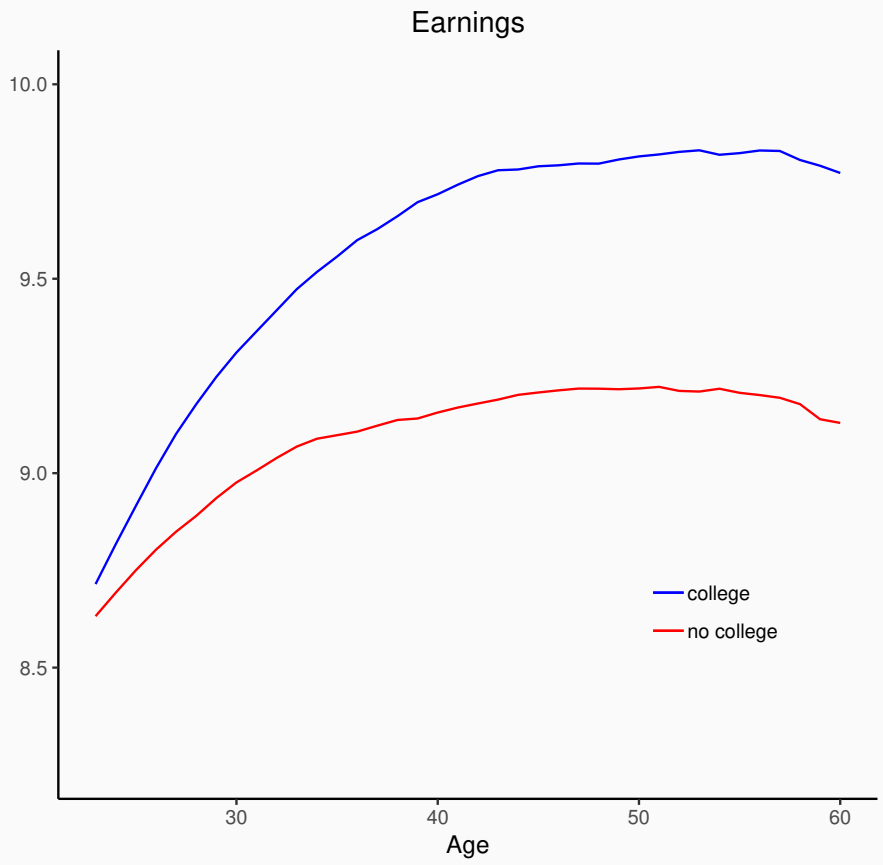

Figure 2: Income profile by education level: PSID

Intergenerational persistence of education

- Information about parent's education.
- Years 1974-2015.
- Focus on individuals aged 23-27.
- Probability of attending college, conditioning on parent's education.

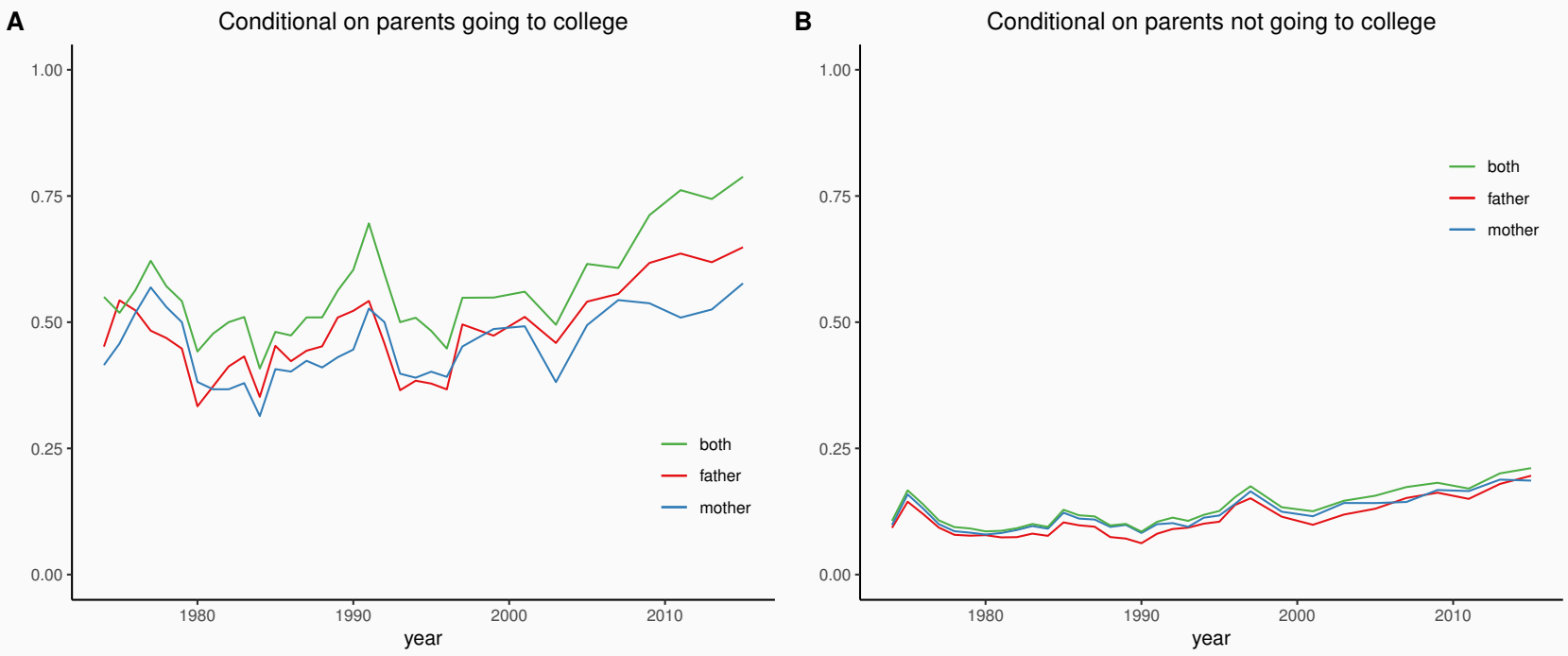

Figure 3: Probability of attending college, conditional on parent's

Education and wealth

- Wealth data from PSID. Years 1984, 1989, 1994 y biannual waves between 1999 y 2015.
- Construct three different wealth classes (bottom, middle, top), according to their relative positions with respect to the rest of individuals in the same cohort.
- Probability of attending college, conditional on wealth class.
- Results are robust to the inclusion of housing in the definition of wealth.

Education and wealth

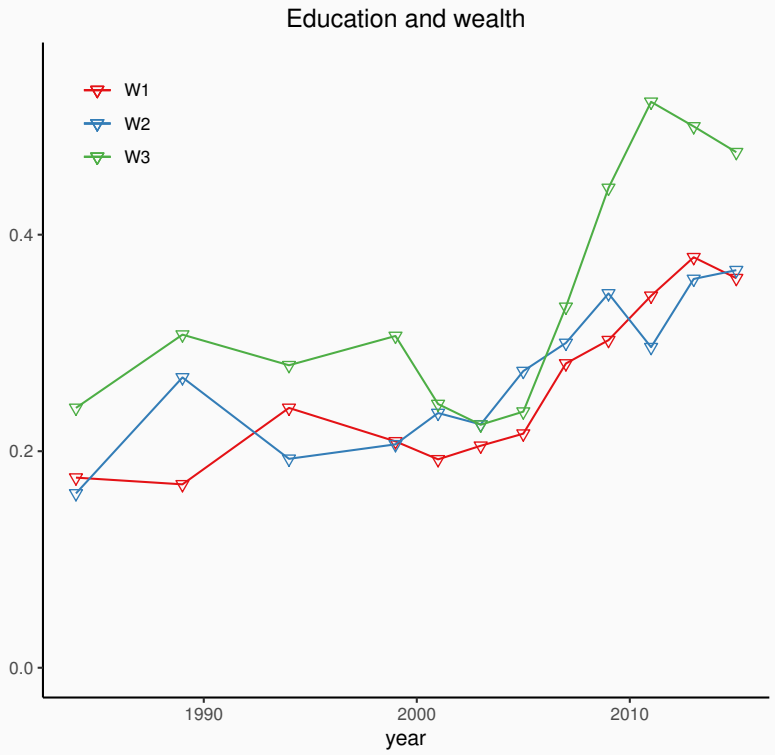

Figure 4: Probability of attending college conditional on wealth class

Education and parental transfers

- Data from NLSY97, cohorts 1980 and 1981.
- Descriptive Statistics.
- Construct “total transfers from parents” as all the transfers made from parents to kids during ages 16 and 22, without considering allowances. Estimates of transfers when the actual sum is not available.
- Results from three models: 1) Linear probability model, 2) Logit model.

Education and parental transfers: descriptive statistics

Cohort	Q1	Median	Mean	Q3	Fraction Tr >0
1980	300	750	3493	2200	39.4%
1981	300	850	3292	2138	39.9%

Table 1: Descriptive statistics transfers

Education and parental transfers: Histogram

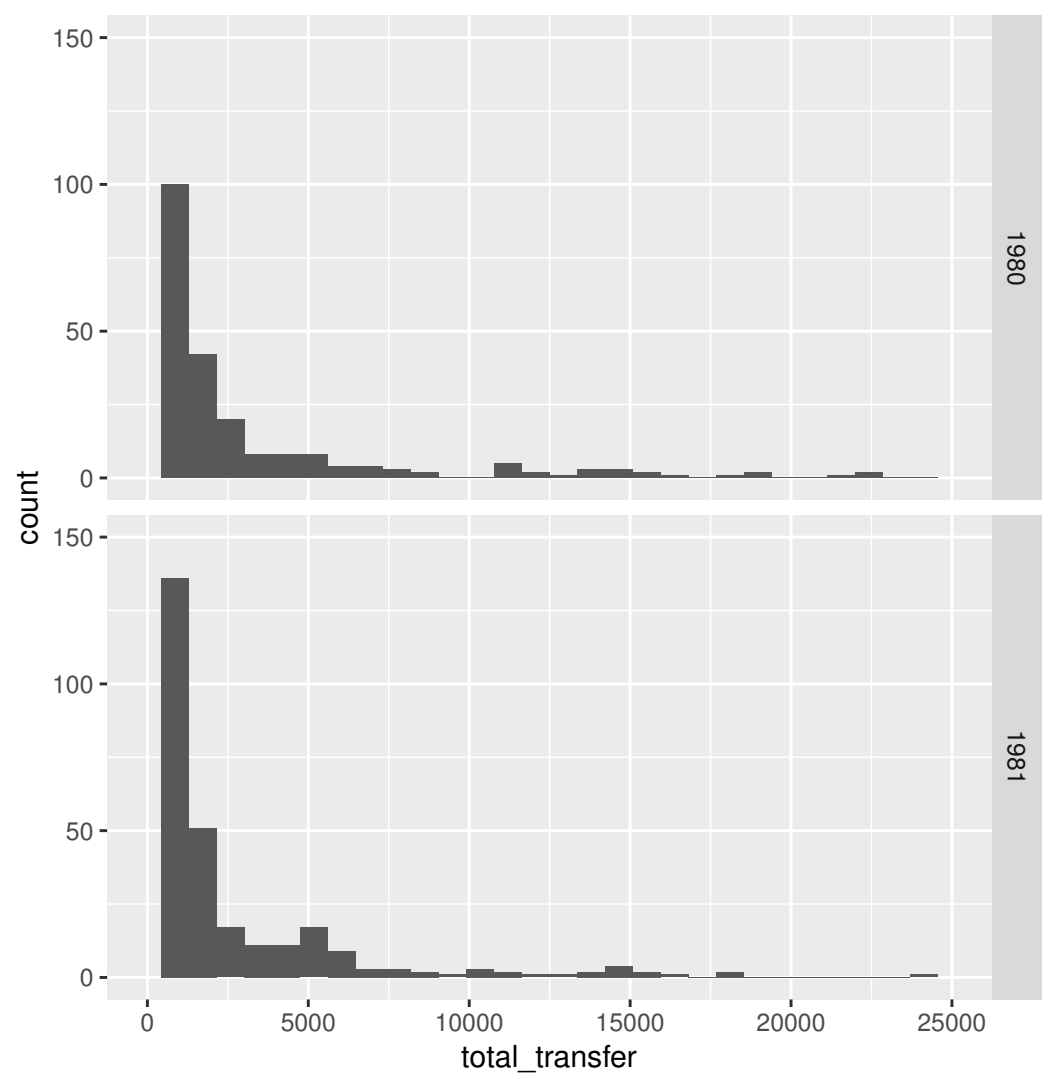

Figure 5: Histogram of positive total transfers

Education and parental transfers: Logit

	Dependent variable:			
	educ			
	(1)	(2)	(3)	(4)
received	1.019***	0.994***	0.757***	0.927***
	(0.104)	(0.105)	(0.121)	(0.140)
gender		0.461***	0.611***	0.601***
		(0.105)	(0.120)	(0.139)
race		0.238***	0.107*	0.091
		(0.051)	(0.060)	(0.076)
log(fam_income)			0.509***	0.363***
			(0.107)	(0.138)
fam_wealth			0.003***	0.003***
			(0.0005)	(0.001)
college_res_p				0.890***
				(0.175)
Constant	−1.288***	−2.791***	−4.834***	−4.462***
	(0.073)	(0.255)	(0.461)	(0.614)
Observations	1,867	1,867	1,635	1,139
Log Likelihood	−1,096.043	−1,075.214	−859.347	−626.856
Akaike Inf. Crit.	2,196.087	2,158.428	1,730.695	1,267.712

Education and parental transfers: Logit - Marginal effects

	Specifications			
	(1)	(2)	(3)	(4)
AME	0.217	0.207	0.140	0.1762
MEM	0.217	0.210	0.154	0.2015

Table 3: Marginal Effects

Calibration

Calibration

Parameter	Meaning	Value	Source/Target
	Prices		
r	rate of return	4%	calibrated
w_l	low wage	1	normalization
w_h	high wage	1.6	(Goldin and Katz (2007))
p	pension	37.5	40% of mean earnings
P_c	price of college education	117.17	share college educated agents (30%)
	Preferences and human capital		
β	Discount Factor	0.95	Standard
σ	CRRA utility function	2	Standard
α	production function	0.7	HVY (2011)
γ	Parameter of Altruism	0.075	(calibrated)
δ	Human capital depreciation	0.0025	(calibrated)
σ_z	Variance human capital shocks	0.047	(calibrated)
	Government		
μ	Tax constant	0.902	Guner et. al (2014)
τ	Tax progressivity	0.036	Guner et. al (2014)
τ_{ss}	Social security tax	0.106	HVT(2011)
	Distribution		
$(\mu_h, \mu_\theta, \sigma_h, \sigma_\theta, \rho_{h\theta})$	initial distribution (log normal)	(90, 017, 56, 0.06, 0.43)	calibrated

Table 4: Calibración

Model Fit: Income Profile

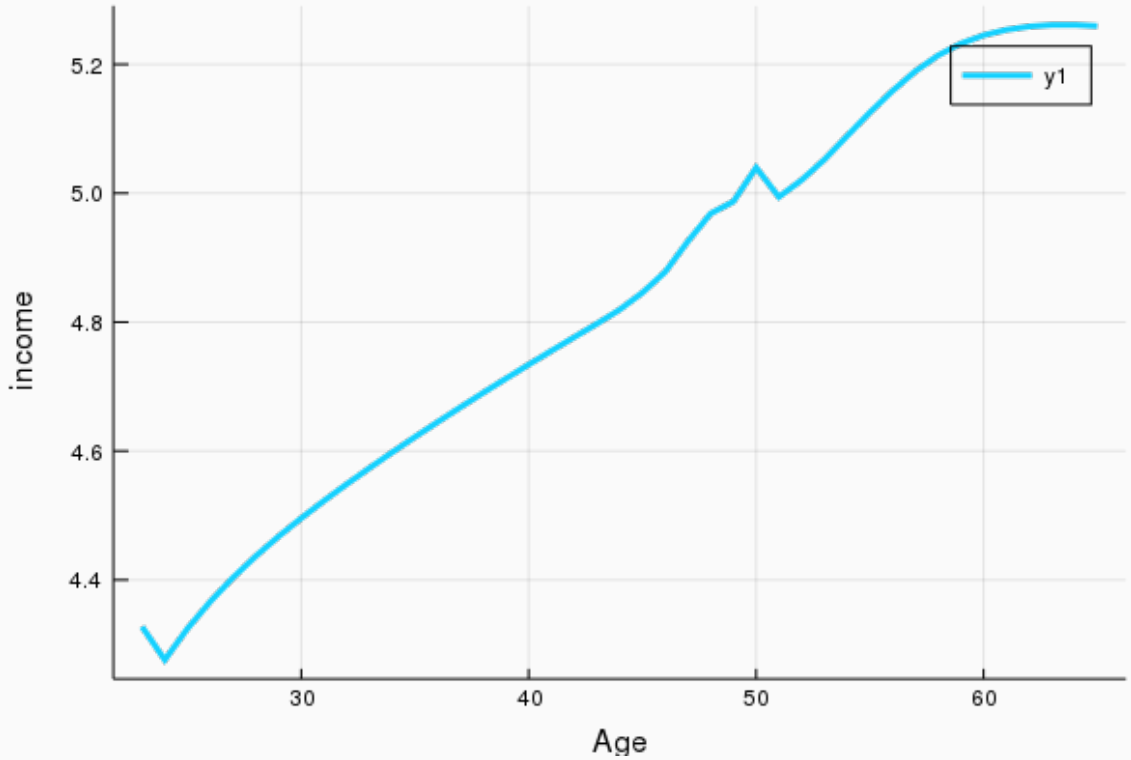

Figure 6: Model income profile

Model Fit: Income Profile by education

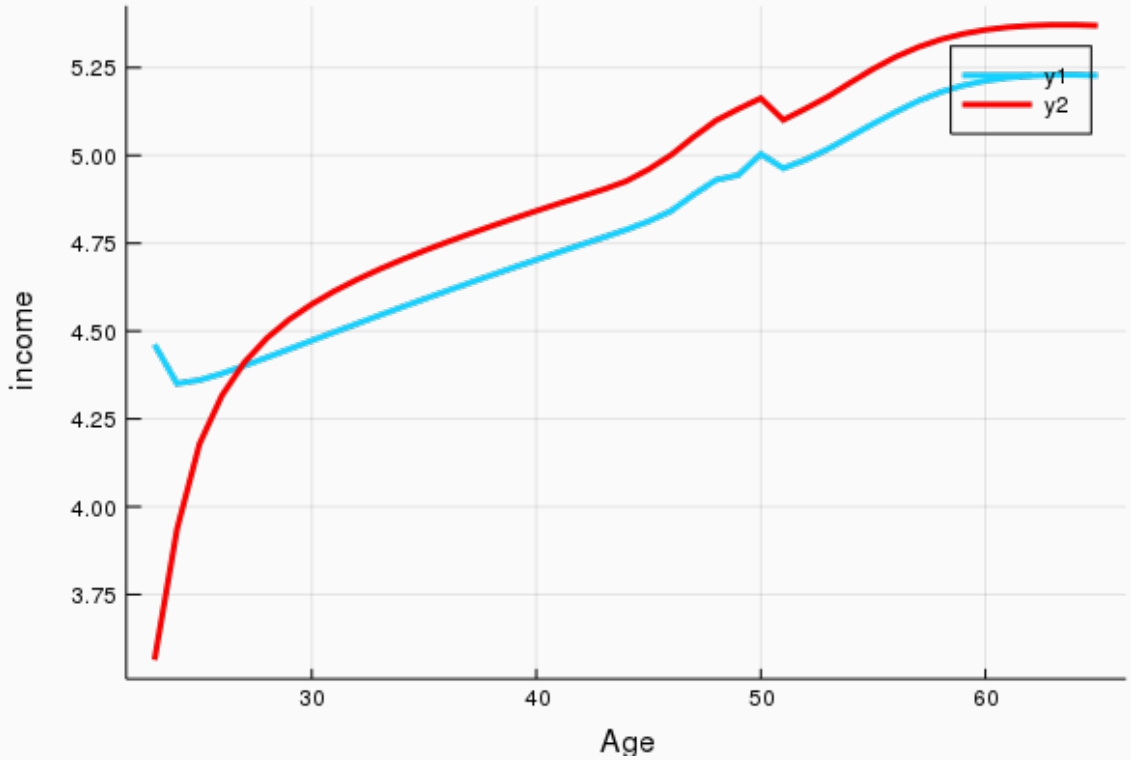

Figure 7: Model income profile

Model Fit: Gini

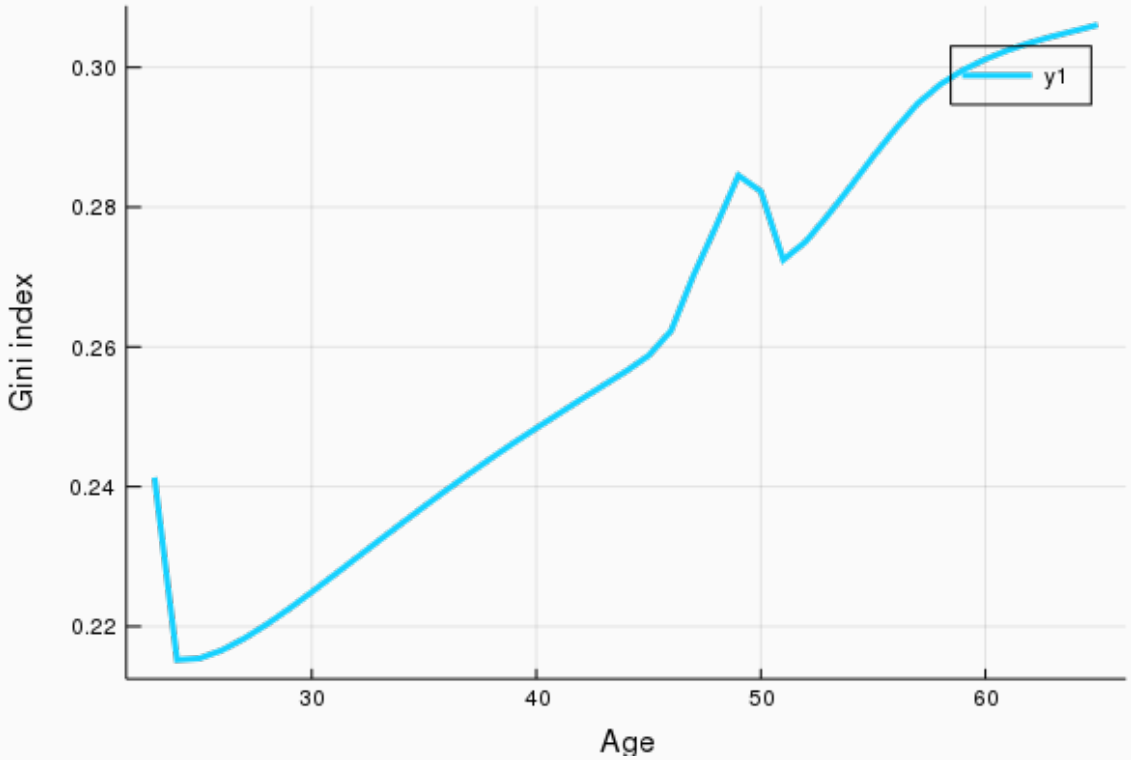

Figure 8: Model Gini index

Model Fit: Initial wealth/ transfer

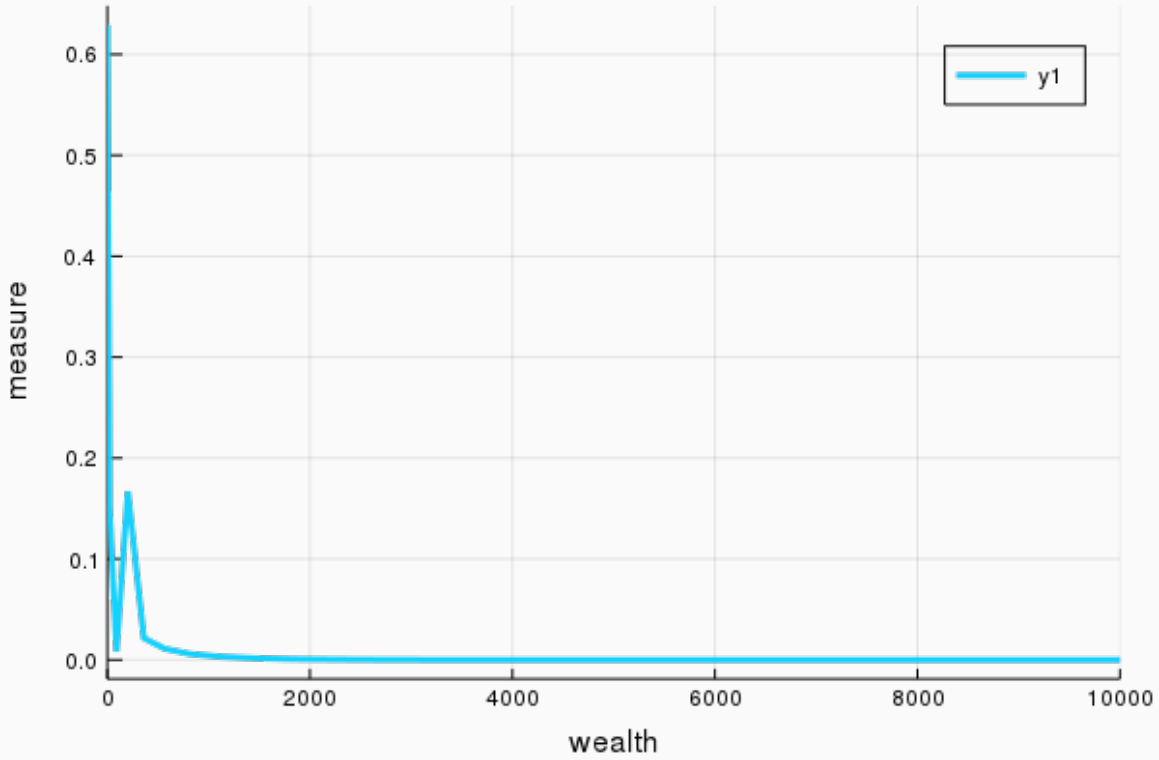

Figure 9: Initial wealth

Model fit: Other moments

- Share of college students: 21.2%
- Share of people with positive transfers: 38%
- Replacement rate in retirement: 30%

Results and Discussion

Sources of Lifetime Inequality

(a, h, θ)	Lifetime utility	Lifetime earnings	Lifetime wealth
(a_m, h_l, θ_m)	-36.5%	-36%	-36%
(a_m, h_h, θ_m)	51.6%	53.5%	53.5%
(a_m, h_m, θ_l)	-2.4%	-7.18%	-7.18%
(a_m, h_m, θ_h)	2.5%	7.18%	7.18%
(a_h, h_m, θ_m)	3.8%	1.7%	3.02%

Table 5: Lifetime Inequality

Variance Decomposition in Dynamic Programming

The present discounted value of utility can be written as

$$B = \sum_{t=0}^{T} \beta^t u(x_t) \tag{1}$$

where x_t is the state at time t, which may follow a Markov process. The value function is define as the sum of expected discounted future pay offs, given the initial state x

$$J(x) = \mathbb{E}(B|x_0 = x) \tag{2}$$

It is useful to define the following operator

$$M(x) = \mathbb{E}(B^2|x_0 = x) \tag{3}$$

Variance Decomposition in Dynamic Programming (2)

Notice that $J(x)$ and $M(x)$ can be written as

$$J(x) = u(x) + \beta P(x'|x)J(x') \tag{4}$$

$$M(x) = u(x)^2 + 2\beta u(x)P(x'|x) + \beta^2 P(x'|x)M(x') \tag{5}$$

The previous two equations can be computed recursively. Using $J(x)$ and $M(x)$ we can compute the variance decomposition of a variable

$$\mathrm{Var}(Y|x) = \mathbb{E}(Y^2|x) - \mathbb{E}(Y|x)^2 = M(x) - J(x)^2 \tag{6}$$

$$\mathrm{var}(Y) = \underbrace{\mathbb{E}(\mathrm{var}(Y|X))}_{\text{due to shocks}} + \underbrace{\mathrm{var}(\mathbb{E}(Y|X))}_{\text{due to initial conditions}} \tag{7}$$

Variance Decomposition

Y	percentage of variance due to initial conditions
Lifetime income	26.5%
Lifetime wealth	26.4%
Lifetime utility	11%

Table 6: Variance decomposition

The Effect of borrowing constraint

(a, h, θ)	Lifetime utility	Lifetime earnings	Lifetime wealth
(a_m, h_l, θ_m)	-37.3%	-34.7%	-34.5%
(a_m, h_h, θ_m)	51.4%	51.2%	50.8%
(a_m, h_m, θ_l)	-3.1%	-9%	-8.9%
(a_m, h_m, θ_h)	3.3%	9%	8.9%
(a_h, h_m, θ_m)	21.1%	26.7%	31.5%

Table 7: Lifetime Inequality

The Effect of borrowing constraint

Y	percentage of variance due to initial conditions
Lifetime income	26.2%
Lifetime wealth	26.1%
Lifetime utility	4.5%

Table 8: Variance decomposition

The Effect of parental background

(a, h, θ)	Lifetime utility	Lifetime earnings	Lifetime wealth
(a_m, h_l, θ_m)	-36.5%	-36%	-36%
(a_m, h_h, θ_m)	51.6%	53.5%	53.5%
(a_m, h_m, θ_l)	-2.4%	-7.18%	-7.18%
(a_m, h_m, θ_h)	2.5%	7.18%	7.18%
(a_h, h_m, θ_m)	4.42%	2.03%	3.46%

Table 9: Lifetime Inequality

The Effect of parental background

Y	percentage of variance due to initial conditions
Lifetime income	32.2%
Lifetime wealth	32.2%
Lifetime utility	11.7%

Table 10: Variance decomposition

Conclusions

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- Quantitative macroeconomic model to explain the evolution of cohort inequality over the life cycle.
- It takes into account the interplay of borrowing constraints and parental transfer in the human capital accumulation process.
- Initial conditions shape the observed inequality. About 25% of the variance of lifetime income and wealth is explained by initial conditions.
- Initial wealth inequality affects lifetime welfare and income. Intergenerational correlation of abilities is less important.